

NET512

Designing and Implementing a Secure, Scalable, and Automated Enterprise Network

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Week 4: The Cisco Troubleshooting Methodology (10 Marks)

22. Attempt to reach the DNS server from a client device and confirm it fails

```
R3#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

23. Work through the OSI layers systematically (Physical → Data Link → Network → Transport) to isolate where the fault lies. Document the checks you perform and the results at each layer.

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

R3#ping r2.net512.local
Translating "r2.net512.local"...domain server (10.10.10.10)
% Unrecognized host or address or protocol not running.
```

24. Identify the root cause of the connectivity failure

```
R2#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.10.10.0/24 is directly connected, FastEthernet0/0
L       10.10.10.2/32 is directly connected, FastEthernet0/0
C       10.10.30.0/24 is directly connected, FastEthernet0/1
L       10.10.30.2/32 is directly connected, FastEthernet0/1

R2#show ip interface brief
^
% Invalid input detected at '^' marker.

R2#show ip interface brief
^
% Invalid input detected at '^' marker.

R2#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.10.10.2      YES manual up              up
FastEthernet0/1          10.10.30.2      YES manual up              up
Vlan1                    unassigned      YES unset  administratively down down
R2#
```

25. Apply the fix.

```
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface fastethernet 0/1
R2(config-if)#ip address 10.10.20.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#
```

```
R3>enable
R3#config t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 10.10.10.10 255.255.255.0 10.10.20.2
%Inconsistent address and mask
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#
```

26. Re-verify connectivity to the DNS server and confirm the fault is resolved.

```
R3#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/1/2 ms
R3#
```
